Elements of Information Theory **2025-2026**

Lecture 2: Method of types

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2.1 Types, type classes

Let $\mathcal{X}$ be a finite alphabet with $|\mathcal{X}| = M$, and consider a sequence $x^n \in \mathcal{X}^n$. The **type** of x^n is its **empirical pmf**, i.e.,

$$\hat{\mathrm{P}}_{x^n}(a) := \frac{1}{n}\left|\{i : x_i = a\}\right| = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{x_i = a\},\ a \in \mathcal{X}.$$

Thus, $\hat{\mathrm{P}}_{x^n} := [\hat{\mathrm{P}}_{x^n}(a) : a \in \mathcal{X}]$ satisfies all the properties of a pmf. Similarly, we can define the **joint type** $\hat{\mathrm{P}}_{x^n,y^n}$ of (x^n, y^n) in $\mathcal{X}^n \times \mathcal{Y}^n$ by considering the couple (a, b) as a symbol, i.e.,

$$\hat{\mathrm{P}}_{x^n,y^n}(a,b) := \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{x_i = a\}\mathbf{1}\{y_i = b\},\ a \in \mathcal{X}, b \in \mathcal{Y}.$$

We can verify that

$$\sum_{b\in\mathcal{Y}} \hat{\mathrm{P}}_{x^n,y^n}(a,b) = \hat{\mathrm{P}}_{x^n}(a), \quad \forall a \in \mathcal{X}$$
$$\sum_{a\in\mathcal{X}} \hat{\mathrm{P}}_{x^n,y^n}(a,b) = \hat{\mathrm{P}}_{y^n}(b), \quad \forall b \in \mathcal{Y}$$

In words, $\hat{\mathrm{P}}_{x^n,y^n}$ is a joint pmf with marginal pmfs $\hat{\mathrm{P}}_{x^n}$ and $\hat{\mathrm{P}}_{y^n}$.

We say that $\hat{\mathrm{P}}$ is a type in $\mathcal{X}^n$ if it is a type of some sequence $x^n \in \mathcal{X}^n$, i.e., there exist $(n_1, \cdots, n_M) \in \mathbb{Z}_+^M$ with $n_1 + \cdots + n_M = n$ and $\hat{\mathrm{P}} = \left[\frac{n_1}{n}, \ldots, \frac{n_M}{n}\right]$. The **set of types** in $\mathcal{X}^n$ is denoted by $\mathcal{P}_n^{\mathcal{X}}$ (or simply $\mathcal{P}_n$). Specifically, we define

$$\mathcal{P}_n^{\mathcal{X}} := \left\{\left[\frac{n_1}{n}, \ldots, \frac{n_M}{n}\right] : \quad n_1 + \cdots + n_M = n, \quad n_i \in \mathbb{Z}_+, i = 1, \ldots, M\right\}$$

The set of types $\mathcal{P}_n^{\mathcal{X}}$ is a finite grid in the **probability simplex**

$$\mathcal{P}^{\mathcal{X}} := \left\{[p_1, \ldots, p_M] \in \mathbb{R}_+^M : \quad p_1 + \cdots + p_M = 1\right\}$$

All sequences with the same type form an equivalent class called **type class**. Specifically, the type class corresponding to a type $\hat{\mathrm{P}}$ is defined as

$$\mathcal{T}^{(n)}(\hat{\mathrm{P}}) := \{x^n \in \mathcal{X}^n :\ \hat{\mathrm{P}}_{x^n}(a) = \hat{\mathrm{P}}(a),\ \forall a \in \mathcal{X}\}.$$

2.2 Size and probability measure of type classes

In the following, we are interesting in finding out

- the number of type classes
- the size of each type class
- the probability of each type class, under a given probability measure

2.2.1 Number of type classes

We can show that the number of types is

$$\boxed{K_{n,M} := |\mathcal{P}_n^{\mathcal{X}}| = \binom{n+M-1}{M-1} \leq (n+1)^{M-1},}$$

since for each of the $M = |\mathcal{X}|$ components in a type $\hat{\mathrm{P}} \in \mathcal{P}_n$ we can at most have $n+1$ possible values, and only $M-1$ of the n_i's are free.

2.2.2 Size of each type class

If it is understood that each type class is defined for a given length n, we can remove the superscript for brevity. The size of each type class is

$$\boxed{|\mathcal{T}(\hat{\mathrm{P}})| = \binom{n}{n_1, \ldots, n_M} := \frac{n!}{\prod_{a\in\mathcal{X}}(n\hat{\mathrm{P}}(a))!}, \quad \hat{\mathrm{P}} \in \mathcal{P}_n}$$

Indeed, for any sequence $x^n \in \mathcal{T}(\hat{\mathrm{P}})$, the set of all $n!$ permutations can be partitioned according to the sequence after permutation. We can check that there are exactly $\prod_{a\in\mathcal{X}}(n\hat{\mathrm{P}}(a))!$ permutations that can transform x^n to $\tilde{x}^n \in \mathcal{T}(\hat{\mathrm{P}})$. Since there are exactly $|\mathcal{T}(\hat{\mathrm{P}})|$ different sequences $\tilde{x}^n$, we must have $n! = |\mathcal{T}(\hat{\mathrm{P}})| \prod_{a\in\mathcal{X}}(n\hat{\mathrm{P}}(a))!$

Let $\hat{\mathrm{P}}, \hat{\mathrm{Q}} \in \mathcal{P}_n$, then

$$\frac{|\mathcal{T}(\hat{\mathrm{P}})|}{|\mathcal{T}(\hat{\mathrm{Q}})|} \geq 2^{n(H(\hat{\mathrm{P}})-H(\hat{\mathrm{Q}}\|\hat{\mathrm{P}}))}$$

In particular, if $\hat{\mathrm{P}}$ is uniform, then $|\mathcal{T}(\hat{\mathrm{P}})| \geq |\mathcal{T}(\hat{\mathrm{Q}})|$ for any $\hat{\mathrm{Q}}$.

Proof. From $|\mathcal{T}(\hat{\mathrm{P}})| = \frac{n!}{(n\hat{\mathrm{P}}(1))!\,(n\hat{\mathrm{P}}(2))!\cdots(n\hat{\mathrm{P}}(M))!}$ and $|\mathcal{T}(\hat{\mathrm{Q}})| = \frac{n!}{(n\hat{\mathrm{Q}}(1))!\,(n\hat{\mathrm{Q}}(2))!\cdots(n\hat{\mathrm{Q}}(M))!}$, we

have

$$\begin{aligned}\frac{|\mathcal{T}(\hat{\mathrm{P}})|}{|\mathcal{T}(\hat{\mathrm{Q}})|} &= \prod_{m=1}^{M} \frac{(n\hat{\mathrm{Q}}(m))!}{(n\hat{\mathrm{P}}(m))!} \\ &\geq \prod_{m=1}^{M} (n\hat{\mathrm{P}}(m))^{n(\hat{\mathrm{Q}}(m)-\hat{\mathrm{P}}(m))} \qquad \left(\tfrac{s!}{t!} \geq t^{s-t} \text{ for all } s,t \in \mathbb{Z}^+\right) \\ &= 2^{n(H(\hat{\mathrm{P}})-H(\hat{\mathrm{Q}}\|\hat{\mathrm{P}}))}. \end{aligned} \tag{2.1}$$

If $\hat{\mathrm{P}}$ is uniform, then $H(\hat{\mathrm{P}}) - H(\hat{\mathrm{Q}}\|\hat{\mathrm{P}}) = \log M - \log M = 0$ and we have $\frac{|\mathcal{T}(\hat{\mathrm{P}})|}{|\mathcal{T}(\hat{\mathrm{Q}})|} \geq 1$. □

2.2.3 Probability of type classes

Let us use the notation P^n or Q^n to denote some product measure. It should be understood that

$$\mathrm{P}^n(x^n) = \prod_{i=1}^{n} \mathrm{P}(x_i), \quad x^n \in \mathcal{X}^n,$$

where P is a pmf[1] defined in $\mathcal{X}$. In words, $\mathrm{P}^n(x^n)$ is the probability of the sequence x^n under the assumption that $x_1, \ldots, x_n$ are i.i.d. $\sim$ P. Such a notation makes the distribution of the random variables involved explicit.

If $x^n \in \mathcal{T}(\hat{\mathrm{P}})$, then its probability under the distribution Q^n, (i.e., the symbols are i.i.d. $\sim$ Q), is exactly

$$\boxed{\mathrm{Q}^n(x^n) = 2^{-nH(\hat{\mathrm{P}}\|\mathrm{Q})}, \quad \forall\, x^n \in \mathcal{T}(\hat{\mathrm{P}}).} \tag{2.2}$$

Or, we can concisely write

$$\boxed{\mathrm{Q}^n(x^n) = 2^{-nH(\hat{\mathrm{P}}_{x^n}\|\mathrm{Q})}}$$

Now we see that under product pmf, sequences inside the same type class have the same probability, i.e., uniformly distributed inside each class. In particular, set $\mathrm{Q} = \hat{\mathrm{P}}$, we have from (2.2.3)

$$\boxed{\hat{\mathrm{P}}^n(x^n) = 2^{-nH(\hat{\mathrm{P}})}, \quad \forall\, x^n \in \mathcal{T}(\hat{\mathrm{P}}).} \tag{2.3}$$

Using $H(\mathrm{P}\|\mathrm{Q}) \geq H(\mathrm{P})$, we have the following.

$$\boxed{\hat{\mathrm{P}}^n(x^n) \geq \mathrm{Q}^n(x^n), \quad \forall\, x^n \in \mathcal{T}(\hat{\mathrm{P}})}$$

In words, a sequence of type $\hat{\mathrm{P}}$ has a larger pmf under the distribution $\hat{\mathrm{P}}^n$ than under any other distribution. However, under the same distribution $\hat{\mathrm{P}}^n$, a sequence of type $\hat{\mathrm{P}}$ does not necessarily has a pmf larger than a sequence outside of the type class. Namely, let $x^n \in \mathcal{T}(\hat{\mathrm{P}})$ and $\tilde{x}^n \in \mathcal{X}^n$ an arbitrary sequence, then

$$\hat{\mathrm{P}}^n(x^n) \not\geq \hat{\mathrm{P}}^n(\tilde{x}^n).$$

[1]Here, we use the same notation for the probability measure and the pmf, which are the same for the discrete case.

The message is that the most typical one is not necessarily the most probable one. Nevertheless, the conclusion is different if we consider the probability of an entire type class. For any Q, let $\mathrm{Q}^n(\mathcal{T}(\hat{\mathrm{P}}))$ be the probability of the set of all the squences in the type class P under the measure Q^n, i.e.,

$$\mathrm{Q}^n(\mathcal{T}(\hat{\mathrm{P}})) := \mathrm{Q}^n(\Big\{x^n :\ x^n \in \mathcal{T}(\hat{\mathrm{P}})\Big\}) = \sum_{x^n \in \mathcal{T}(\hat{\mathrm{P}})} \mathrm{Q}^n(x^n).$$

We also call it the Q^n-probability of the type class $\hat{\mathrm{P}}$. Since each sequence has the same probability, it is easily seen that

$$\mathrm{Q}^n(\mathcal{T}(\hat{\mathrm{P}})) = |\mathcal{T}(\hat{\mathrm{P}})| 2^{-nH(\hat{\mathrm{P}}\|\mathrm{Q})} \tag{2.4}$$

$$\hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}})) = |\mathcal{T}(\hat{\mathrm{P}})| 2^{-nH(\hat{\mathrm{P}})} \tag{2.5}$$

$$\begin{aligned} &\hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}})) \geq \mathrm{Q}^n(\mathcal{T}(\hat{\mathrm{P}})), \quad \forall \hat{\mathrm{P}} \in \mathcal{P}_n, \mathrm{Q} \in \mathcal{P} \\ &\hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}})) \geq \hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{Q}})), \quad \forall \hat{\mathrm{P}}, \hat{\mathrm{Q}} \in \mathcal{P}_n \end{aligned} \tag{2.6}$$

In words, the probability of an entire type class is larger under the same distribution than under any other distribution. Furthermore, under the distribution $\hat{\mathrm{P}}^n$, the type class $\mathcal{T}(\hat{\mathrm{P}})$ has a higher probability than any other type classes.

Proof. The first inequality is straightforward from (2.2.3), (2.2.3), and $H(\mathrm{P}\|\mathrm{Q}) \geq H(\mathrm{P})$.

To prove the second one, taking the ratio, we have

$$\frac{\hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}}))}{\hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{Q}}))} = \frac{|\mathcal{T}(\hat{\mathrm{P}})|}{|\mathcal{T}(\hat{\mathrm{Q}})|} 2^{-n(H(\hat{\mathrm{P}}) - H(\hat{\mathrm{Q}}\|\hat{\mathrm{P}}))}. \tag{2.7}$$

From (2.2.2) and (2.2.3), we show that $\frac{\hat{\mathrm{Q}}^n(\mathcal{T}(\hat{\mathrm{Q}}))}{\hat{\mathrm{Q}}^n(\mathcal{T}(\hat{\mathrm{P}}))} \geq 1$. □

2.2.4 Size and probability of the type classes, revisited

Remarkably, the probability of the type classes can help us to obtain bounds on the size of each type class, although the latter does not depend on any probability distribution.

The size of each type class is bounded as

$$(n+1)^{-|\mathcal{X}|} 2^{nH(\hat{\mathrm{P}})} \leq |\mathcal{T}(\hat{\mathrm{P}})| \leq 2^{nH(\hat{\mathrm{P}})}, \quad \hat{\mathrm{P}} \in \mathcal{P}_n.$$

For brevity, we can write[a]

$$|\mathcal{T}(\hat{\mathrm{P}})| \doteq 2^{nH(\hat{\mathrm{P}})}.$$

[a] where $\doteq$ means equality in exponent in the large n regime, i.e., $f(n) \doteq g(n)$ means

$$\lim_{n\to\infty} \frac{\log f(n)}{n} = \lim_{n\to\infty} \frac{\log g(n)}{n}.$$

Proof. The upper bound is straightforward from (2.2.3)

$$1 \geq \hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}})) = |\mathcal{T}(\hat{\mathrm{P}})| 2^{-nH(\hat{\mathrm{P}})}.$$

The lower bound can be obtained using (2.2.3):

$$\begin{aligned} 1 &= \sum_{\hat{\mathrm{Q}} \in \mathcal{P}_n} \hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{Q}})) \\ &\leq \sum_{\hat{\mathrm{Q}} \in \mathcal{P}_n} \hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}})) \\ &= |\mathcal{P}_n| \hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}})) \\ &\leq (n+1)^{|\mathcal{X}|} \hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}})) \\ &\leq (n+1)^{|\mathcal{X}|} |\mathcal{T}(\hat{\mathrm{P}})| 2^{-nH(\hat{\mathrm{P}})} \end{aligned}$$

□

The main message is that the size of a type class of $\hat{\mathrm{P}}$ is roughly $2^{nH(\hat{\mathrm{P}})}$, up to a polynomial factor in n. The size increases with the entropy given by the type. Intuitively, the more uniform the type, the larger the type class.

Then, as a simple consequence of the size bounds, combined with (2.2.3), we can have the following bounds on the probability measure of a given type class $\mathcal{T}(\hat{\mathrm{P}})$ under the distribution Q^n.

$$\boxed{(n+1)^{-|\mathcal{X}|} 2^{-nD(\hat{\mathrm{P}}\|\mathrm{Q})} \leq \mathrm{Q}^n(\mathcal{T}(\hat{\mathrm{P}})) \leq 2^{-nD(\hat{\mathrm{P}}\|\mathrm{Q})}}$$

implying

$$\mathrm{Q}^n(\mathcal{T}(\hat{\mathrm{P}})) \doteq 2^{-nD(\hat{\mathrm{P}}\|\mathrm{Q})}.$$

In particular, we have

$$\hat{\mathrm{P}}^n(\mathcal{T}(\hat{\mathrm{P}})) \geq (n+1)^{-|\mathcal{X}|}.$$

Intuitively, the probability of generating a sequence with a mismatched type $\mathrm{Q} \neq \hat{\mathrm{P}}$ decreases exponentially with n as $2^{-nD(\hat{\mathrm{P}}\|\mathrm{Q})}$, while the probability of generating a squence with a matched type decreases only polynomially with n not faster than $(n+1)^{-|\mathcal{X}|}$.

2.3 Strongly typical sequences

For each pmf P and $\varepsilon \in [0,1]$, we define the **strongly typical set**, or ε-typical n-sequences set, as

$$\begin{aligned} \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}) &:= \{x^n : \quad |\hat{\mathrm{P}}_{x^n}(a) - \mathrm{P}(a)| \leq \varepsilon \mathrm{P}(a),\ \forall\, a \in \mathcal{X}\} \\ &= \{x^n : \quad (1-\varepsilon)\mathrm{P}(a) \leq \hat{\mathrm{P}}_{x^n}(a) \leq (1+\varepsilon)\mathrm{P}(a),\ \forall\, a \in \mathcal{X}\} \end{aligned}$$

Thus, the typical set contains sequences with types "close" to the given pmf P. In particular, when $\varepsilon = 0$, we have $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}) = \mathcal{T}(\mathrm{P})$ if P is a type and $\emptyset$ otherwise. Sometimes, we write $\mathcal{T}_\varepsilon^{(n)}(\mathrm{X})$ instead of $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$ if $\mathrm{X} \sim \mathrm{P}$. Whenever confusion is unlikely, we can simply write $\mathcal{T}_\varepsilon^{(n)}$, or $\mathcal{T}_\varepsilon$.

The following **typical average lemma** shows the power of strong typicality.

For any function $g : \mathcal{X} \to \mathbb{R}$, and $\mathrm{X} \sim \mathrm{P}$

$$\mathbb{E}g(\mathrm{X}) - \delta(\varepsilon) \le \frac{1}{n}\sum_{i=1}^{n} g(x_i) \le \mathbb{E}g(\mathrm{X}) + \delta(\varepsilon), \quad \forall\, x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}),$$

where $\delta(\varepsilon) := \varepsilon\mathbb{E}|g(\mathrm{X})|$. Equivalently, we write

$$|\mathbb{E}_{\mathrm{P}} g(\mathrm{X}) - \mathbb{E}_{\hat{\mathrm{P}}_{x^n}} g(\mathrm{X})| \le \varepsilon \mathbb{E}_{\mathrm{P}}|g(\mathrm{X})|, \quad \forall\, x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}),$$

Typical sequences have the following properties.

1. **Sequence.** For every $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$,

$$2^{-n(1+\varepsilon)H(\mathrm{P})} \le \mathrm{P}^n(x^n) \le 2^{-n(1-\varepsilon)H(\mathrm{P})}, \tag{2.8}$$

and

$$D(\hat{\mathrm{P}}_{x^n}\|\mathrm{P}) \le \delta_D(\varepsilon) := \log(1+\varepsilon)$$
$$|H(\hat{\mathrm{P}}_{x^n}) - H(\mathrm{P})| \le \delta_H(\varepsilon, \mathrm{P}) := \varepsilon H(\mathrm{P}) - \log(1-\varepsilon)$$

2. **Probability of the set.** The probability of the typical set

$$\lim_{n\to\infty} \mathrm{P}^n(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})) = 1, \quad \text{if } \lim_{n\to\infty} n\varepsilon^2 = \infty. \tag{2.9}$$

More specifically, we have

$$\mathrm{P}^n(\mathrm{X}^n \notin \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})) \le \delta_T(\varepsilon, n, \mathrm{P}) := 2M^* e^{-2n\varepsilon^2 c_{\mathrm{P}}} \tag{2.10}$$

where $c_P := \min_{a\in\mathcal{X}:\mathrm{P}(a)>0} \mathrm{P}(a)^2$ and $M^* := |\{a \in \mathcal{X} : \mathrm{P}(a) > 0\}|$.

3. **Size of the set.** The size of the typical set is bounded by

$$2^{n(1+\varepsilon)H(\mathrm{P})} \ge |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| \ge 2^{n(1-\varepsilon)H(\mathrm{P}) - \delta'_T(\varepsilon,n,\mathrm{P})},$$

where

$$\delta'_T(\varepsilon, n, \mathrm{P}) := -\log(1 - \delta_T(\varepsilon, n, \mathrm{P}))^+ \tag{2.11}$$

that goes to 0 whenever $n\varepsilon^2 \to \infty$.

Proof. 1. The bounds (1) are straightforward from the typical average lemma where we let $g(x) = \log \mathrm{P}(x)$. And we apply the lemma on $\log \mathrm{P}^n(x^n) = \sum_i g(x_i)$. Note that $\mathbb{E}|\log \mathrm{P}(X)| = -\mathbb{E}\log \mathrm{P}(X) = H(\mathrm{P})$. The divergence bound is from the fact that $\hat{\mathrm{P}} \ll \mathrm{P}$ and that $\log \frac{\hat{\mathrm{P}}(a)}{\mathrm{P}(a)} \le \log(1+\varepsilon)$. Taking the expectation, we obtain the upper bound. Finally, for the entropy bounds, we write

$$H(\hat{\mathrm{P}}) = \mathbb{E}_{\hat{\mathrm{P}}} \log \frac{1}{\hat{\mathrm{P}}(\mathrm{X})}$$

$$\leq \mathbb{E}_{\hat{\mathrm{P}}} \log \frac{1}{\mathrm{P(X)}} + \log \frac{1}{1-\varepsilon}$$
$$\leq (1+\varepsilon)\mathbb{E}_{\mathrm{P}} \log \frac{1}{\mathrm{P(X)}} + \log \frac{1}{1-\varepsilon}$$
$$H(\hat{\mathrm{P}}) \geq \mathbb{E}_{\hat{\mathrm{P}}} \log \frac{1}{\mathrm{P(X)}} - \log \frac{1}{1+\varepsilon}$$
$$\leq (1-\varepsilon)\mathbb{E}_{\mathrm{P}} \log \frac{1}{\mathrm{P(X)}} - \log(1+\varepsilon)$$

Therefore, $|H(\hat{\mathrm{P}}) - H(\mathrm{P})| \leq \varepsilon H(\mathrm{P}) + \max\{\log(1+\varepsilon), -\log(1-\varepsilon)\} = \varepsilon H(\mathrm{P}) - \log(1-\varepsilon)$.

2. We check the probability

$$\mathrm{P}^n(\mathrm{X}^n \notin \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})) = \mathrm{P}^n\left(\bigcup_{a\in\mathcal{X}} \left\{\left|\frac{1}{n}\sum_{i=1}^n \mathbf{1}\{\mathrm{X}_i = a\} - \mathrm{P}(a)\right| > \varepsilon \mathrm{P}(a)\right\}\right) \tag{2.12}$$
$$\leq \sum_{a\in\mathcal{X}} \mathrm{P}^n\left(\left\{\left|\frac{1}{n}\sum_{i=1}^n \mathbf{1}\{\mathrm{X}_i = a\} - \mathrm{P}(a)\right| > \varepsilon \mathrm{P}(a)\right\}\right) \tag{2.13}$$
$$= \sum_{a\in\mathcal{X},\mathrm{P}(a)>0} \mathrm{P}^n\left(\left\{\left|\frac{1}{n}\sum_{i=1}^n \mathbf{1}\{\mathrm{X}_i = a\} - \mathrm{P}(a)\right| > \varepsilon \mathrm{P}(a)\right\}\right) \tag{2.14}$$
$$\leq \sum_{a\in\mathcal{X},\mathrm{P}(a)>0} 2e^{-2n\varepsilon^2 \mathrm{P}(a)^2}, \tag{2.15}$$

where in (2.3), we use the fact that the event has probability 0 when $\mathrm{P}(a) = 0$; the last inequality is from Hoeffding's inequality.

> **Hoeffding's inequality** Let $\mathrm{X}_1, \ldots, \mathrm{X}_n$ be independent random variables such that $\mathrm{X}_i \in [a_i, b_i]$ almost surely. Define $\mathrm{S}_n := \frac{1}{n}\sum_{i=1}^n \mathrm{X}_i$ and $\Delta_n := \frac{1}{n}\sum_{i=1}^n (a_i - b_i)^2$. We have
>
> $$\mathbb{P}\{|\mathrm{S_n} - \mathbb{E}(\mathrm{S}_n)| \geq t\} \leq 2e^{-\frac{2nt^2}{\Delta_n}}. \tag{2.16}$$

Specifically, we have $\mathbf{1}\{\mathrm{X}_1 = a\}, \ldots, \mathbf{1}\{\mathrm{X}_n = a\}$ i.i.d., bounded in $[0,1]$, and with expectation $\mathrm{P}(a)$, so the average concentrates around the mean exponentially fast. Replacing each term in the sum by the dominant one, i.e., when $\mathrm{P}(a)$ is the smallest, we get the upper bound (2). (2) follows directly.

3. The upper bound is straightforward from $1 \geq \mathrm{P}^n(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})) \geq 2^{-n(1+\varepsilon)H(\mathrm{P}))}|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})|$. From (2), $\mathrm{P}^n(\mathrm{X}^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})) \geq (1 - \delta_T(\varepsilon, n, \mathrm{P}))^+$. Since we also have $\mathrm{P}^n(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})) \leq |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})|\, 2^{-n(1-\varepsilon)H(\mathrm{P})}$, we prove that $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| 2^{-n(1-\varepsilon)H(\mathrm{P})} \geq (1 - 2|\mathcal{X}|e^{-2n\varepsilon^2 c_{\mathrm{P}}})^+$ which proves the lower bound on the size of the typical set.

□

From the above properties, we see that for a memoryless stationary source P^n, the probability mass is concentrated on a small subset $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$ with approximately $2^{nH(\mathrm{P})}$ sequences, out of the set of all $M^n = 2^{n\log M}$ sequences. Moreover, every sequence (generated by the source P^n) inside $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$ has approximately the same probability $2^{-nH(\mathrm{P})}$. This is also known as the **asymptotic equipartition property (AEP)**.

2.4 Jointly typical sequences, conditional typical sequences

In the exact same way, we can define the **jointly typical sequences** with respect to the joint pmf P_{XY}.

$$\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}) := \{(x^n, y^n) : \quad |\hat{\mathrm{P}}_{x^n,y^n}(a,b) - \mathrm{P}_{XY}(a,b)| \le \varepsilon \mathrm{P}_{XY}(a,b),\ \forall\, a \in \mathcal{X}, b \in \mathcal{Y}\}$$

We also define the **conditionally typical sequences** with respect to the joint pmf P_{XY}.

$$\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY} \,|\, y^n) := \{x^n : \quad (x^n, y^n) \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})\}.$$

By definition, we have

$$\boxed{(x^n, y^n) \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}) \quad \Leftrightarrow \quad x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY} \,|\, y^n) \quad \Leftrightarrow \quad y^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY} \,|\, x^n)}$$

We have the following properties. [2]

[2] With a slight abuse of notation, by $\mathrm{P}_X^n(x^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{X})}$ we mean $\mathrm{P}_X^n(x^n) \in [2^{-n(1+\varepsilon)H(\mathrm{X})}, 2^{-n(1-\varepsilon)H(\mathrm{X})}]$.

1. If $(x^n, y^n) \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})$, then
   - $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_X),\ y^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_Y)$
   - $\mathrm{P}_X^n(x^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{X})}$, $\mathrm{P}_Y^n(y^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{Y})}$, $\mathrm{P}_{XY}^n(x^n, y^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{X},\mathrm{Y})}$
   - $\mathrm{P}_{Y|X}^n(y^n|x^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{Y}\,|\,\mathrm{X})}$, $\mathrm{P}_{X|Y}^n(x^n|y^n) \approx 2^{-n(1\pm\varepsilon)H(\mathrm{X}\,|\,\mathrm{Y})}$
2. $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})| \le 2^{n(1+\varepsilon)H(\mathrm{P}_{XY})}$, and $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})| \ge 2^{n(1-\varepsilon)H(\mathrm{P}_{XY})-\delta_T(\varepsilon,n,\mathrm{P}_{XY})}$
3. For every $y^n \in \mathcal{Y}^n$,
$$|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)| \le 2^{n(1+\varepsilon)H(\mathrm{X}|\mathrm{Y})}.$$
4. **Conditional typicality lemma**. Let $\mathrm{P}_{X^n|Y^n} = \mathrm{P}_{X|Y}^n$. For every $y^n \in \mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_Y)$,
$$\lim_{n\to\infty} \mathrm{P}_{X^n|Y^n=y^n}(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)) = 1, \quad \text{if } \lim_{n\to\infty} n\varepsilon^2 = \infty.$$
Specifically, we have
$$\mathrm{P}_{X^n|Y^n=y^n}(\mathrm{X}^n \notin \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)) \le 2|\mathcal{X}|\,|\mathcal{Y}|e^{-n\varepsilon^2 c_P/2},$$
where $c_P := \min_{a,b:\mathrm{P}_{XY}(a,b)>0} \mathrm{P}_{XY}(a,b)^2$.
5. Let $y^n \in \mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_Y)$. Then,
$$|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)| \ge 2^{n(1-\varepsilon)H(\mathrm{X}|\mathrm{Y})-\delta'_T(\varepsilon/2,n,\mathrm{P}_{XY})}. \tag{2.17}$$
where δ'_T is defined in (3).
6. **Joint typicality lemma**.
   - For every $y^n \in \mathcal{Y}^n$,
$$\mathrm{P}_X^n(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)) \le 2^{-n(I(\mathrm{X};\mathrm{Y})-2\varepsilon H(\mathrm{X}))}. \tag{2.18}$$
   - Let $y^n \in \mathcal{T}_{\varepsilon/2}^{(n)}$. Then,
$$\mathrm{P}_X^n(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)) \ge 2^{-n(I(\mathrm{X};\mathrm{Y})+2\varepsilon H(\mathrm{X}))-\delta'_T(\varepsilon/2,n,\mathrm{P}_{XY})}. \tag{2.19}$$

Proof. 1. The first two items are straightforward from the definition of jointly typical sequences. Marginalizing, we verify that the sequences x^n and y^n are individually typical with the same ε. To prove the third item, we need to apply the typical average lemma to P_{XY} and the function $g(x,y) = \log \mathrm{P}_{X|Y}(x|y)$, using the fact that $\log(\mathrm{P}_{X^n|Y^n}(x^n\,|\,y^n)) = \sum_{i=1}^n \log \mathrm{P}_{X|Y}(x_i\,|\,y_i)$.

2. Same as for typical sequences. (cf. previous section)

3. Since $\mathrm{P}_{X|Y}^n(x^n|y^n) \geq 2^{-n(1+\varepsilon)H(\mathrm{X}|\mathrm{Y})}$ from the first property, we have

$$\begin{aligned} 1 &\geq \sum_{x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)} \mathrm{P}_{X|Y}^n(x^n|y^n) && (2.20)\\ &\geq \sum_{x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)} 2^{-n(1+\varepsilon)H(\mathrm{X}|\mathrm{Y})} && (2.21)\\ &= |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)| 2^{-n(1+\varepsilon)H(\mathrm{X}|\mathrm{Y})}. && (2.22) \end{aligned}$$

4. It is enough to show that for any $(a,b) \in \mathcal{X} \times \mathcal{Y}$

$$\mathrm{P}_{X^n|Y^n=y^n}(|\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b) - \mathrm{P}_{XY}(a,b)| > \varepsilon \mathrm{P}_{XY}(a,b)) \to 0 \tag{2.23}$$

and apply the union bound to finish the proof since $|\mathcal{X}\times\mathcal{Y}|$ is finite. When $\mathrm{P}_{XY}(a,b) = 0$, the above condition is verified trivially. In the following, we focus on the case where $\mathrm{P}_{XY}(a,b) > 0$, which is equivalent to $\mathrm{P}_X(a) > 0$, $\mathrm{P}_Y(b) > 0$, and $\mathrm{P}_{X|Y}(a|b) > 0$. Note that

$$\pi_{\mathrm{X}^n,y^n}(a,b) = \frac{1}{n}\sum_{i=1}^n \mathbf{1}\{(\mathrm{X}_i, y_i) = (a,b)\} \tag{2.24}$$

is a sum of n independent random variables. Indeed, X_i's are independent, each generated according to the distribution $\mathrm{P}_{X_i|Y_i=y_i}$. This implies that $\mathbf{1}\{(\mathrm{X}_i, y_i) = (a,b)\}$ are independent for different i's. The expectation of average is $\mathbb{E}_{\mathrm{P}_{X^n|Y^n=y^n}}\left(\frac{1}{n}\sum_{i=1}^n \mathbf{1}\{(\mathrm{X}_i, y_i) = (a,b)\}\right) = \hat{\mathrm{P}}_{y^n}(b)\mathrm{P}_{X|Y}(a|b)$. Due to the triangle inequality, we have

$$\begin{aligned} &|\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b) - \mathrm{P}_{XY}(a,b)| && (2.25)\\ &\leq |\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b) - \hat{\mathrm{P}}_{y^n}(b)\mathrm{P}_{X|Y}(a\,|\,b)| + |\hat{\mathrm{P}}_{y^n}(b)\mathrm{P}_{X|Y}(a\,|\,b) - \mathrm{P}_{XY}(a,b)| && (2.26)\\ &\leq |\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b) - \hat{\mathrm{P}}_{y^n}(b)\mathrm{P}_{X|Y}(a\,|\,b)| + \frac{\varepsilon}{2}\mathrm{P}_{XY}(a,b), && (2.27) \end{aligned}$$

where we use the fact that $(1-\varepsilon/2)\mathrm{P}_Y(b) \leq \hat{\mathrm{P}}_{y^n}(b) \leq (1+\varepsilon/2)\mathrm{P}_Y(b)$ to establish the last inequality. Now, we can bound the probability

$$\begin{aligned} &\mathrm{P}_{X^n|Y^n=y^n}(|\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b) - \mathrm{P}_{XY}(a,b)| > \varepsilon \mathrm{P}_{XY}(a,b)) && (2.28)\\ &\leq \mathrm{P}_{X^n|Y^n=y^n}(|\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b) - \hat{\mathrm{P}}_{y^n}(b)\mathrm{P}_{X|Y}(a|b)| > (\varepsilon - \varepsilon/2)\mathrm{P}_{XY}(a,b)) && (2.29)\\ &\leq 2e^{-n\varepsilon^2 \mathrm{P}_{XY}(a,b)^2/2} && (2.30) \end{aligned}$$

according to Hoeffding's inequality (2). From the union bound, we have

$$\begin{aligned} \mathrm{P}_{X^n|Y^n=y^n}(\mathrm{X}^n \notin \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)) &\leq \sum_{a,b:\mathrm{P}_{XY}(a,b)>0} \mathrm{P}_{X^n|Y^n=y^n}(|\hat{\mathrm{P}}_{\mathrm{X}^n,y^n}(a,b) - \mathrm{P}_{XY}(a,b)| > \varepsilon \mathrm{P}_{XY}(a,b))\\ &\leq 2|\mathcal{X}|\,|\mathcal{Y}| e^{-2n\varepsilon^2 c_P/2}, \end{aligned} \tag{2.31}$$

where $c_P := \min_{a,b:\mathrm{P}_{XY}(a,b)>0} P_{XY}(a,b)^2$. Note that this bound goes to 0 when $n\varepsilon^2 \to \infty$. Hence, as long as $n\varepsilon^2 \to \infty$, we have the condition (4).

5. On the one hand, from the conditional typicality, we have $\mathrm{P}_{X^n|Y^n=y^n}(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)) \geq (1 - 2|\mathcal{X}|\,|\mathcal{Y}|e^{-n\varepsilon^2 c_P/2})^+$. On the other hand, we have $\mathrm{P}_{X^n|Y^n=y^n}(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)) \leq |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,y^n)| 2^{-n(1-\varepsilon)H(\mathrm{X}\,|\,\mathrm{Y})}$. Combining both inequalities, we prove the lower bound (5).

6. To show the upper bound, apply the upper bound on the size of the set $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)| \le 2^{n(1+\varepsilon)H(\mathrm{X}\,|\,\mathrm{Y})}$ and the upper bound on the probability of $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_X)$ (implied by the fact that $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)$) under P_X^n, i.e., $\mathrm{P}_X^n(x^n) \le 2^{-n(1-\varepsilon)H(\mathrm{X})}$. We have

$$\begin{aligned}
\mathrm{P}_X^n(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)) &= \sum_{x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)} \mathrm{P}_X^n(x^n) && (2.32)\\
&\le \sum_{x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)} 2^{-n(1-\varepsilon)H(\mathrm{X})} && (2.33)\\
&\le 2^{n(1+\varepsilon)H(\mathrm{X}\,|\,\mathrm{Y})} 2^{-n(1-\varepsilon)H(\mathrm{X})} && (2.34)\\
&= 2^{-n(I(\mathrm{X};\mathrm{Y})-\varepsilon H(\mathrm{X})-\varepsilon H(\mathrm{X}\,|\,\mathrm{Y}))} && (2.35)\\
&\le 2^{-n(I(\mathrm{X};\mathrm{Y})-2\varepsilon H(\mathrm{X}))}. && (2.36)
\end{aligned}$$

To show the lower bound, apply the lower bound (5) on size of the set $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)$ and the lower bound on the probability of $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_X)$ (implied by the fact that $x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)$) under P_X^n. We have

$$\begin{aligned}
\mathrm{P}_X^n(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)) &= \sum_{x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)} \mathrm{P}_X^n(x^n) && (2.37)\\
&\ge \sum_{x^n \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)} 2^{-n(1+\varepsilon)H(\mathrm{X})} && (2.38)\\
&\ge 2^{n(1-\varepsilon)H(\mathrm{X}\,|\,\mathrm{Y})-\delta_T(\varepsilon/2,n,\mathrm{P}_{XY})} 2^{-n(1+\varepsilon)H(\mathrm{X})} && (2.39)\\
&\ge 2^{-n(I(\mathrm{X};\mathrm{Y})+2\varepsilon H(\mathrm{X}))-\delta_T(\varepsilon/2,n,\mathrm{P}_{XY})}. && (2.40)
\end{aligned}$$

□

2.5 Covering lemma, packing lemma

From the joint typicality lemma, we can derive the following covering lemma and packing lemma that are useful for proving source coding and channel coding theorems, respectively.

Covering lemma. Let $\mathrm{X}^n(m) \sim \mathrm{P}_X^n$ for $m = 1, \ldots, 2^{nR_n}$ be mutually independent and also independent of some Y^n. Then,

$$\lim_{n\to\infty} \mathbb{P}\left\{ \bigcup_{m=1,\ldots,2^{nR_n}} \left\{ (\mathrm{X}^n(m), \mathrm{Y}^n) \in \mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}) \right\} \right\} = 1,$$

if the following conditions are satisfied:[a]

$$\begin{aligned}
\lim_{n\to\infty} n\varepsilon^2 &= \infty && (2.41)\\
\lim_{n\to\infty} \mathbb{P}\left\{ \mathrm{Y}^n \in \mathcal{T}_{\varepsilon/2}(\mathrm{P}_Y) \right\} &= 1 && (2.42)\\
\lim_{n\to\infty} nR_n - nI(\mathrm{X};\mathrm{Y}) - 2n\varepsilon H(\mathrm{X}) &= \infty. && (2.43)
\end{aligned}$$

[a] We don't require $\mathrm{Y}^n \sim \mathrm{P}_Y^n$. Instead, we only need that Y^n is typical with respect to P_Y with high probability.

Proof.

$$\mathbb{P}\left\{\bigcup_{m=1,\ldots,2^{nR_n}}\left\{(\mathrm{X}^n(m),\mathrm{Y}^n)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})\right\}\right\}=\sum_{y^n}\mathrm{P}_{Y^n}(y^n)\mathbb{P}\left\{\bigcup_{m=1,\ldots,2^{nR_n}}\left\{(\mathrm{X}^n(m),y^n)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})\right\}\right\}$$

$$\geq\sum_{y^n\in\mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_Y)}\mathrm{P}_{Y^n}(y^n)\mathbb{P}\left\{\bigcup_{m=1,\ldots,2^{nR_n}}\left\{(\mathrm{X}^n(m),y^n)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})\right\}\right\}$$

$$=\sum_{y^n\in\mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_Y)}\mathrm{P}_{Y^n}(y^n)\mathbb{P}\left\{\bigcup_{m=1,\ldots,2^{nR_n}}\left\{\mathrm{X}^n(m)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)\right\}\right\}$$

For any given y^n, the events in the unions in the second probability are independent. From the probability lower bound (6), we have the following upper bound for $y^n\in\mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_Y)$,

$$\mathbb{P}\left\{\mathrm{X}^n(m)\notin\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)\right\}\leq 1-2^{-n(I(\mathrm{X};\mathrm{Y})+2\varepsilon H(\mathrm{X}))-\delta_T'(\varepsilon/2,n,\mathrm{P}_{XY})}.$$

Applying the upper bound $1-x\leq e^{-x}$, and the independence between $\mathrm{X}^n(m)$, we have

$$\mathbb{P}\left\{\bigcap_{m=1,\ldots,2^{nR_n}}\left\{\mathrm{X}^n(m)\notin\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}|y^n)\right\}\right\}\leq\exp\left(-2^{nR}2^{-n(I(\mathrm{X};\mathrm{Y})+2\varepsilon H(X))-\delta_T'(\varepsilon/2,n,\mathrm{P}_{XY})}\right) \tag{2.44}$$

$$=\exp\left(-2^{n(R_n-I(\mathrm{X};\mathrm{Y})-2\varepsilon H(\mathrm{X}))-\delta_T'(\varepsilon/2,n,\mathrm{P}_{XY})}\right) \tag{2.45}$$

Therefore, we have

$$\mathbb{P}\left\{\bigcup_{m=1,\ldots,2^{nR_n}}\left\{(\mathrm{X}^n(m),\mathrm{Y}^n)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})\right\}\right\}\geq\sum_{y^n\in\mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_Y)}\mathrm{P}_{Y^n}(y^n)\left(1-\exp\left(-2^{n(R_n-I(\mathrm{X};\mathrm{Y})-2\varepsilon H(\mathrm{X}))-\delta_T'(\varepsilon/2,n,\mathrm{P}_{XY}}\right.\right.$$

$$=\mathrm{P}_{Y^n}(\mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_Y))\left(1-\exp\left(-2^{n(R_n-I(\mathrm{X};\mathrm{Y})-2\varepsilon H(\mathrm{X}))-\delta_T'(\varepsilon/2,n,\mathrm{P}_{XY})}\right)\right)$$

From conditions (2.25), (2.26), and (2.27), we have $\delta_T'(\varepsilon/2,n,\mathrm{P}_{XY})\to 0$, $\mathrm{P}_{Y^n}(\mathcal{T}_{\varepsilon/2}^{(n)}(\mathrm{P}_Y))\to 1$, and $n(R_n-I(\mathrm{X};\mathrm{Y})-2\varepsilon H(\mathrm{X}))\to\infty$. It follows that the above probability lower bound goes to 1. □

Packing lemma. Let $\mathrm{X}^n(m)\sim\mathrm{P}_X^n$ for $m=1,\ldots,2^{nR_n}$ (not necessarily independent for different m) be independent of an arbitrary random sequence Y^n. Then,

$$\lim_{n\to\infty}\mathbb{P}\left\{\bigcup_{m=1,\ldots,2^{nR_n}}\left\{(\mathrm{X}^n(m),\mathrm{Y}^n)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})\right\}\right\}=0,$$

if

$$\lim_{n\to\infty}nR_n-nI(\mathrm{X};\mathrm{Y})+2n\varepsilon H(\mathrm{X})=-\infty.$$

Proof. It is enough to apply the union bound and the probability upper bound (6). Indeed, from the union bound, we have

$$\mathbb{P}\left\{\bigcup_{m=1,\dots,2^{nR_n}}\left\{(\mathrm{X}^n(m),\mathrm{Y}^n)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY})\right\}\right\}=\mathbb{P}\left\{\bigcup_{m=1,\dots,2^{nR_n}}\left\{\mathrm{X}^n(m)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,\mathrm{Y}^n)\right\}\right\} \tag{2.46}$$

$$\leq\sum_{m=1}^{2^{nR_n}}\mathbb{P}\left\{\mathrm{X}^n(m)\in\mathcal{T}_\varepsilon^{(n)}(\mathrm{P}_{XY}\,|\,\mathrm{Y}^n)\right\} \tag{2.47}$$

$$\leq 2^{nR_n-nI(\mathrm{X};\mathrm{Y})+2n\varepsilon H(\mathrm{X})}, \tag{2.48}$$

where the last inequality is from the joint typicality lemma (6). The upper bound goes to 0 if

$$\lim_{n\to\infty} nR_n-nI(\mathrm{X};\mathrm{Y})+2n\varepsilon H(\mathrm{X})=-\infty.$$

□

Exercises

1. Types and type classes by hand. Let $\mathcal{X} = \{a, b, c\}$ and $n = 3$.

   - List the types in $\mathcal{P}_3$ and check that $K_{3,3} = \binom{5}{2}$. Compare with the bound $(n+1)^{M-1}$.
   - Give the size of each type class, and check that the sizes add up to $|\mathcal{X}^3| = 27$.
   - Let $\mathrm{Q} = \left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$. Compute $\mathrm{Q}^3(aab)$ directly, then with the formula $\mathrm{Q}^n(x^n) = 2^{-nH(\hat{\mathrm{P}}_{x^n} \| \mathrm{Q})}$.
   - Compute $\mathrm{Q}^3\big(\mathcal{T}^{(3)}(\hat{\mathrm{P}})\big)$ for every $\hat{\mathrm{P}} \in \mathcal{P}_3$, and check that the probabilities add up to 1. Which sequence is the most likely under Q^3? Which type classes are the most likely?

2. Number of types [CK 2.1]. Let $M = |\mathcal{X}|$.

   - Show that $K_{n,M} = |\mathcal{P}_n| = \binom{n+M-1}{M-1}$. *Hint: a type is given by the counts $(n_1, \dots, n_M)$. Write n stars on a line and put $M-1$ bars among them to split the stars into M groups; for instance, $\star\star\,|\,|\star$ stands for $(2, 0, 1)$.*
   - Show that $K_{n,M} \le (n+1)^{M-1}$ without computing $K_{n,M}$. *Hint: the last count is fixed by the others.*
   - Compute $K_{n,2}$ and $K_{4,3}$, and compare with the bound.

3. Size of a type class. Let $\mathcal{X} = \{a_1, \dots, a_M\}$, let $\hat{\mathrm{P}} \in \mathcal{P}_n$ with counts $n_i := n\hat{\mathrm{P}}(a_i)$, and let $x^n, \tilde{x}^n \in \mathcal{T}^{(n)}(\hat{\mathrm{P}})$.

   - Show that exactly $\prod_{i=1}^M n_i!$ permutations σ of $\{1, \dots, n\}$ satisfy $x_{\sigma(k)} = \tilde{x}_k$ for all k. *Example: $x^4 = 0110$ and $\tilde{x}^4 = 1100$.*
   - Deduce that
   $$|\mathcal{T}^{(n)}(\hat{\mathrm{P}})| = \frac{n!}{\prod_{i=1}^M n_i!} = \binom{n}{n_1, \dots, n_M}.$$
   Hint: count the $n!$ permutations of x^n according to the sequence they produce.
   - Deduce the multinomial identity $\sum \binom{n}{n_1, \dots, n_M} = M^n$, where the sum is over $(n_1, \dots, n_M) \in \mathbb{Z}_+^M$ with $\sum_i n_i = n$.

4. Maximum-likelihood probability of a type class. For $\hat{\mathrm{P}} \in \mathcal{P}_n$, recall that $\gamma_n(\hat{\mathrm{P}}) := \hat{\mathrm{P}}^n\big(\mathcal{T}^{(n)}(\hat{\mathrm{P}})\big) = |\mathcal{T}^{(n)}(\hat{\mathrm{P}})|\, 2^{-nH(\hat{\mathrm{P}})}$.

   - Let $\mathcal{X} = \{0, 1\}$. Show that $\gamma_n\big(\left[\frac{k}{n}, 1 - \frac{k}{n}\right]\big) = \binom{n}{k}\big(\frac{k}{n}\big)^k\big(1 - \frac{k}{n}\big)^{n-k}$, compute it for $n = 4$ and $k = 0, \dots, 4$, and check that $K_{4,2}^{-1} \le \gamma_4 \le 1$.
   - Let $\tilde{M}$ be the size of the support of $\hat{\mathrm{P}}$, and U the uniform distribution on it. With the Stirling approximation of $|\mathcal{T}^{(n)}(\hat{\mathrm{P}})|$ from the lecture, show that, for a fixed $\hat{\mathrm{P}}$, as $n \to \infty$ over the n for which $\hat{\mathrm{P}} \in \mathcal{P}_n$,
   $$\gamma_n(\hat{\mathrm{P}}) \simeq (2\pi n)^{-\frac{\tilde{M}-1}{2}}\, \tilde{M}^{\frac{\tilde{M}}{2}}\, 2^{\frac{\tilde{M}}{2} D(\mathrm{U} \| \hat{\mathrm{P}})}.$$
   Hint: write $n_1 \cdots n_{\tilde{M}} = n^{\tilde{M}} \prod_{a:\, \hat{\mathrm{P}}(a) > 0} \hat{\mathrm{P}}(a)$ and
   $$-\frac{1}{2} \sum_{a:\, \hat{\mathrm{P}}(a) > 0} \log \hat{\mathrm{P}}(a) = \frac{\tilde{M}}{2}\big(\log \tilde{M} + D(\mathrm{U} \| \hat{\mathrm{P}})\big).$$

- Deduce that $\gamma_n([p, 1-p]) \simeq \frac{1}{\sqrt{2\pi np(1-p)}}$ for $0 < p < 1$, and compare with the values for $n = 4$.
- Among the types with support size $\tilde{M}$, which one has the smallest γ_n for large n? Is its type class large or small?

5. Probability of a rare event. Let $\mathcal{X} = \{0, 1\}$, $\mathrm{P} = [0.8,\ 0.2]$ (so that $\mathrm{P}(1) = 0.2$), $\mathrm{X}^n \sim \mathrm{P}^n$, and let $\mathcal{A}_n$ be the set of sequences x^n with at least $\frac{n}{2}$ ones, i.e., with $\hat{\mathrm{P}}_{x^n}(1) \geq \frac{1}{2}$.

   - Write $\mathrm{P}^n(\mathcal{A}_n)$ as a sum over types, and show that $\mathrm{P}^n(\mathcal{A}_n) \leq (n+1)\, 2^{-nD^*}$, with $D^* := D\big(\big[\frac{1}{2}, \frac{1}{2}\big] \big\| \mathrm{P}\big)$. *Hint: show that $t \mapsto D([1-t, t] \| \mathrm{P})$ is increasing on $[0.2, 1]$.*
   - Show that $\mathrm{P}^n(\mathcal{A}_n) \geq (n+1)^{-1}\, 2^{-nD^*}$ for even n.
   - Conclude that $-\frac{1}{n} \log \mathrm{P}^n(\mathcal{A}_n) \to D^*$, and compute D^*. Why does the sum over types not change the exponent?

6. Universal source coding. To encode $x^n \in \mathcal{X}^n$ with bits, send first the index of its type $\hat{\mathrm{P}}_{x^n}$ in $\mathcal{P}_n$, then the index of x^n in its type class $\mathcal{T}^{(n)}(\hat{\mathrm{P}}_{x^n})$, with fixed-length binary indices of $\lceil \log K_{n,M} \rceil$ and $\lceil \log |\mathcal{T}^{(n)}(\hat{\mathrm{P}}_{x^n})| \rceil$ bits.

   - Explain why the decoder recovers x^n, and why the code does not depend on the distribution of the source.
   - Show that the length $\ell(x^n)$ of the code word of x^n satisfies $\ell(x^n) \leq nH(\hat{\mathrm{P}}_{x^n}) + (M-1)\log(n+1) + 2$.
   - Let $\mathrm{X}^n \sim \mathrm{P}^n$. Show that $\mathbb{E}\big[\hat{\mathrm{P}}_{\mathrm{X}^n}\big] = \mathrm{P}$ and deduce that $\frac{1}{n}\mathbb{E}\big[\ell(\mathrm{X}^n)\big] \leq H(\mathrm{P}) + \frac{(M-1)\log(n+1)+2}{n}$. *Hint: H is concave.*

7. Typical set of a binary source. Let $\mathcal{X} = \{0, 1\}$, $\mathrm{P} = [0.2,\ 0.8]$, $n = 1000$ and $\varepsilon = 0.2$. This problem uses the properties of typical sequences (Section 2.3).

   - Which numbers of zeros k make x^n ε-typical? Express $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})|$ and $\mathrm{P}^n\big(\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})\big)$ as sums over k.
   - Give an upper bound on $\mathrm{P}^n\big(\mathrm{X}^n \notin \mathcal{T}_\varepsilon^{(n)}(\mathrm{P})\big)$.
   - Give lower and upper bounds on $\log |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})|$, first with the properties of the typical set, then with the largest type class contained in it.
   - With a computer, evaluate the sums of the first part and compare.

8. Exact size of the typical set. Let $\mathrm{P} \in \mathcal{P}(\mathcal{X})$ and $\varepsilon > 0$, and define

$$H_\varepsilon(\mathrm{P}) := \max\big\{H(\mathrm{Q}) :\ \mathrm{Q} \in \mathcal{P}(\mathcal{X}),\ |\mathrm{Q}(a) - \mathrm{P}(a)| \leq \varepsilon \mathrm{P}(a),\ \forall\, a \in \mathcal{X}\big\}.$$

   - Using that $\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})$ is a union of type classes, show that $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| \leq K_{n,M}\, 2^{nH_\varepsilon(\mathrm{P})}$, and that $|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| \geq K_{n,M}^{-1}\, 2^{nH(\hat{\mathrm{P}})}$ for every type $\hat{\mathrm{P}}$ that satisfies the constraint. Choosing types $\hat{\mathrm{P}}_n$ that converge to a maximizer, conclude that

$$\lim_{n\to\infty} \frac{1}{n} \log |\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})| = H_\varepsilon(\mathrm{P}).$$

   - Show that $H(\mathrm{P}) \leq H_\varepsilon(\mathrm{P}) \leq (1+\varepsilon)H(\mathrm{P})$, and that $H_\varepsilon(\mathrm{P}) = H(\mathrm{P})$ if and only if P is uniform on its support. *Hint:* $H(\mathrm{Q}) \leq H(\mathrm{Q}\|\mathrm{P}) = \mathbb{E}_{\mathrm{Q}}\big[\log \frac{1}{\mathrm{P}(\mathrm{X})}\big]$.

- Show that a maximizer is

$$\mathrm{Q}^*(a) = \min\bigl\{\max\{c,\ (1-\varepsilon)\mathrm{P}(a)\},\ (1+\varepsilon)\mathrm{P}(a)\bigr\}, \qquad a \in \mathcal{X},$$

  where c is chosen such that $\sum_a \mathrm{Q}^*(a) = 1$: it flattens P toward the uniform distribution as much as the constraint allows. *Hint: H is concave; write the KKT conditions.*

- Compute $H_\varepsilon(\mathrm{P})$ for the binary source of the previous problem. Compare it with $(1-\varepsilon)H(\mathrm{P})$, $H(\mathrm{P})$ and $(1+\varepsilon)H(\mathrm{P})$, and with the value of $\frac{1}{n}\log|\mathcal{T}_\varepsilon^{(n)}(\mathrm{P})|$ computed in the previous problem.